Cell Systems

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Voices

What do you most hope we will achieve with mammalian synthetic biology within the next decade?



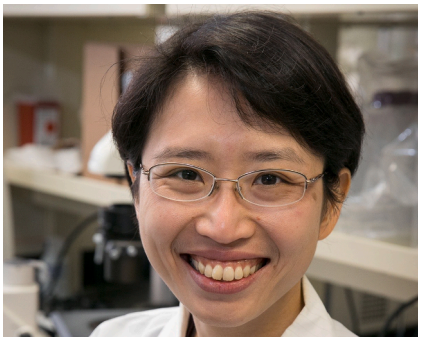
Martin Fussenegger
ETH Zürich

The software layer for medicine

Over the next decade, mammalian synthetic biology should become the first precision control layer for human physiology. Synthetic gene circuits will sense molecules—glucose, lipids, and cytokines—and actuate closed-loop, real-time corrections integrated with host metabolism. This retires open-loop, Paracelsus-era dosing—pills at fixed intervals—in favor of continuous control. Placement should follow safety and access, not organ identity: subcutaneous sensing and control, by autologous cells, exosome/RNA cargoes, or xenogeneic tissue, and encapsulation or immune shielding manage risk. Metabolic control is modular—blood glucose does not need be controlled from the pancreas, nor do lipids need to be lowered by the liver. As circuits interface with metabolism, diagnosis, prevention, and therapy converge—shifting care from chronic management to cure.

Bioelectronic wireless control ports—traceless cues and electrogenetic links—connect circuits to wearables and implants, enabling natural-language control—"Hey Google, control my blood glucose" or "Hey Siri, adjust my blood fat levels"—within guardrails for dose, reversibility, and audit.

Getting there requires moving beyond artisanal builds. AI-orchestrated design-build-test-learn (DBTL) with virtual cell and tissue models and active learning should become routine so that circuits are specified, model-checked, and fail-safe before reaching patients. Done well, everyone gains a programmable, biocompatible control loop—easy to supervise—that maintains health: a software layer for medicine.



Yvonne Y. Chen
University of California, Los Angeles

Democratizing cell therapies

Mammalian synthetic biology has already generated examples of real-world applications such as chimeric antigen receptor (CAR)-T cell therapies for cancer. Looking forward, I am optimistic that mammalian synthetic biology will continue to address practical problems with practical solutions and expand beyond ad hoc cell manufacturing and oncology applications. For example, the ability to understand and exquisitely control cell differentiation from pluripotent stem cells could open up new avenues for allogeneic cell therapy. *In vivo* cell engineering provides another modality by which to manipulate mammalian cells for therapeutic purposes. Beyond oncology, mammalian synthetic biology could play a critical role in addressing significant unmet needs such as neurodegenerative diseases and autoimmune disorders.

A key question is how to make mammalian cell therapies not only efficacious but also safe and scalable for indications that impact millions of patients worldwide. To date, mammalian cell engineering has been characterized by a roughly bimodal distribution. On one end are clinically implemented therapies where the design is relatively minimalistic, with single receptors or simple inducible gene cassettes. On the other end are highly complex genetic circuits that can perform higher-order computation under ideal conditions but are difficult to implement in a practical setting. A major goal for the next ten years would be to engineer systems that can achieve refined functionalities while maintaining system robustness and compatibility with real-world conditions. Greater collaboration between synthetic biologists and clinicians will facilitate such developments, as well as emerging tools in gene editing and cell manufacturing.



Wilson Wong
Boston University

Built to last

Gene silencing is a widely known but rarely discussed problem in synthetic biology. Many complex genetic circuits in my lab failed due to epigenetic silencing. This issue is especially problematic in applications such as tissue engineering, where long-term experimentation is often required. While gene silencing is commonly observed, the reason for this silencing remains poorly defined and appears highly cell and context dependent. Various isolated efforts are being undertaken to try to mitigate this problem, such as avoiding the use of an inducible or synthetic promoter in favor of a regulatable mRNA. While clever, bypassing commonly used promoters is not a sustainable strategy if we wish to leverage the full potential of genetic engineering. Ultimately, we will need to understand why some genetic elements are more prone to silencing than others are. We can address this challenge by collecting more data first. We need to characterize silencing kinetics and epigenetic markers for as many genetic parts as possible across different cell types and genome locations. This dataset will at least provide the much-needed information on the long-term performance of synthetic genetic circuits in various contexts. Furthermore, this dataset could serve as a foundation for training AI/machine learning (ML) models to understand the cause of the silencing and develop solutions to address this challenge.



Nicole Borth
BOKU University Vienna

Mimicking complexity

For mammalian cells, we have, over the last decade, come to better understand the complexity of regulation of many genes that come together to achieve a specific behavior. There is often redundancy in which genes may cooperate to achieve the same goal, and frequently, it is not a shutting down or turning on of any gene but subtle adjustments across a larger number of genes that work together. This complexity and cooperative spirit are difficult to mimic by synthetic biology means—therefore, I hope that can we manage to generate more multiplexed, interactive tools that sense each other's status and respond accordingly. This will require that we learn to understand the autologous regulatory circuits of cells and learn from them to develop and design synthetic biology tools that speak a language that cells already know.



Susan Rosser
University of Edinburgh

Engineering in control

Mammalian synthetic biology is an exciting and very diverse field of research encompassing applications in cell and gene therapies, regenerative medicine, drug discovery, and biomanufacturing. In the next decade, underpinned by progress in AI and automation, my hope is that mammalian synthetic biology will deliver in three areas. First, in cell and gene therapies, mammalian synthetic biology advances will enable safe and effective therapeutics via engineered genetic control systems, where transgenes are expressed in the right cell type and at the right level for the right amount of time. A key need for these areas is the development of new delivery systems, whether biobased, such as new viral vectors, or biocompatible systems, such as engineered nanoparticles. My second hope is that during the next ten years, mammalian synthetic biology will make a step toward change in medicine by engineering "sentinel cells," which via a closed loop control circuit, could recognize disease-related biomarkers, process the information, and then produce an appropriate therapeutic molecule at the correct location and in the correct concentration to treat the disease before it develops or progresses. Third, many of the most expensive drugs on the market are biologics. My hope is that mammalian synthetic biology can be used to engineer production cell lines such that these therapeutics can be produced more cheaply and efficiently, enabling cost reductions for patients and health services and accessibility for the developing world, with subsequent huge benefits economically and socially.



Leonardo Morsut
University of Southern California

Collaborating with biology's agency

In the next decade, I hope mammalian synthetic biology will learn to engage the intrinsic agency of living systems—their capacity to generate and pursue goals such as healing, maintaining form, or building structure. I think about this agency as inextricably linked to the notion of fields—supra-cellular patterns and constraints co-created by cells and their environment, shaped, for example, by morphogen gradients, electrical states, mechanical tensions, glycocalyx landscapes, and histories inscribed in chromatin. I hope we develop synthetic systems that operate at the level of these fields—using new proteins, genetic elements, and complex circuits built in the relevant cells, and supported by computational tools capable of field-level design—so that our interventions work in concert with the system's intrinsic dynamics. Realizing this vision will also require experimental pipelines that measure integrated tissue-level behaviors, giving biological systems space to reveal their goals so that iteration can proceed with those goals in mind rather than as an afterthought. This shift could expand synthetic biology's design space, enabling interventions that harness rather than constrain the system's innate creativity. I don't expect this challenge to be resolved within a decade, but I would be disappointed not to see real movement in this direction. If synthetic biology can align its designs with cellular agency and the guiding structure of morphogenetic fields, we may move from controlling living systems to co-creating with them, opening paths for living systems to generate forms and functions we could never script.



Barbara Di Ventura
University of Freiburg

Cell therapies and sustainable food

Synthetic biologists strive to solve current societal issues such as food shortage, water and soil contamination, carbon emissions, and human diseases. Microbes are, for most applications, the ideal chassis due to their fast growth, available engineering tools, simpler regulation, and resilience in hostile environments. However, for human therapy, engineered human cells are a more natural choice. Engineering eukaryotic cells remains challenging due to their complex endogenous regulation, yet the field of mammalian synthetic biology has already achieved many important successes. In the next decade, I hope to see a shift in the cell types being engineered, moving away from the immortalized lines typically used in mammalian synthetic biology toward primary human cells or induced pluripotent stem cells. I would also like to see more efforts devoted to reducing the potential immunogenicity of engineered components, which could otherwise limit the applicability of cell-based therapies. While a few groups are beginning to address this, most mammalian synthetic biology tools are still built without this aspect in mind, understandably because it is almost impossible by definition to engineer orthogonal control systems using endogenous building blocks. Finally, although cellular agriculture is still in its infancy, I hope to witness significant progress in cultured-meat efforts that harness mammalian synthetic biology. As a potential alternative to animal farming, cultured meat could meaningfully reduce emissions, but it must become truly sustainable to fulfill this promise.



Michael Garton
University of Toronto

Machine design of cell cures

Our terribly incomplete understanding of mammalian cell and molecular mechanisms has, up to now, made the synthetic biologist a mole-eyed and clumsy species. This is now rapidly changing with the advent of generative models—where design constraints can be learned implicitly from evolution and easily integrated with the engineering process. I hope that, over the next decade, the astonishing improvement in the design success rate that machine intelligence affords will accelerate robust solutions to current challenges bedeviling the augmentation of human-cell-based therapy development with synthetic biology. Challenges include robust receptor design, epigenetic silencing, resource loading, component immunogenicity (innate and adaptive), and synthetic circuit—endogenous pathway compatibility. I hope that with efforts focused on generating the right data for training—executed most efficiently with the use of active learning—we can harness the advances in generative AI to develop human cell implants that can sense and respond to disease ebb and flow. All this toward one-and-done quasi-cures can be routinely delivered to patients in clinic ten years from now. I also hope that, instead of eroding our

motivation to understand mechanism, observing the machines at work will provide a slew of new mechanistic hypotheses for the curious scientist to explore.



Joshua N. Leonard
Northwestern University

Toward immune-integrated design

Our immune system learns and enforces self/non-self discrimination. Modulating this process drove advances from vaccines to cancer immunotherapy, but immune rejection limits technologies spanning gene editors, delivery vehicles, and engineered cells. These risks are partially managed by evasion (e.g., removing epitopes recognized by antibodies) and suppression (e.g., systemic immune inhibition). A promising opportunity is immune-integrated design—building synthetic biology technologies for intentional immune coordination from the bottom up.

Converging advances now make this urgent and approachable. We can increasingly predict epitopes that can be presented to prime adaptive immunity. This constraint could guide protein design and re-design powered by AI and computational tools. Induced tolerance can extend beyond exposure therapy to clonal deletion of auto-reactive B and T cells and targeted peripheral tolerance using regulatory T cells. Biodesign is advancing beyond unhelpful dichotomies. Some heuristic categories (e.g., viral/non-viral) are irrelevant. Any non-self protein can be recognized as foreign, including *de novo* designed proteins and human proteins not subjected to central tolerance. Immunogenicity is not unidimensional. Polly Matzinger explained that danger primes non-self responses; context matters. Immune-integrated design of molecules, cells, and tissues could improve efficacy, safety (e.g., containment), and translation of new technologies to patient benefits.



Lacramioara Bintu
Stanford University

Robust gene regulation models and tools

We live in a golden era for gene regulation and mammalian synthetic biology: we can rapidly build DNA constructs and deliver them to cells to reprogram their function. This is exciting because synthetic biology can answer fundamental questions in biology and biophysics while also enabling new genetic and cellular medicines. Yet once inside cells, synthetic constructs still behave in ways we cannot fully anticipate. For example, synthetic promoters and proximal enhancers or silencers often yield all-or-none outputs—ON in some cells and OFF in others—likely reflecting chromatin-mediated regulation. Transgene silencing is another striking example, with constructs unexpectedly shutting down depending on delivery mode, genomic context, differentiation trajectory, or cellular environment. We clearly need better models and tools to predict and control gene expression levels and dynamics across cell types and signaling states.

To robustly predict and control expression, we need a deeper understanding of how synthetic constructs function within the dynamic chromatin landscape. Gene regulation is tightly linked to recruitment of chromatin regulators, including complexes that remodel nucleosomes and deposit histone and DNA modifications. Incorporating RNA-level regulation is also essential, given growing evidence of crosstalk between RNA and chromatin. To generalize input-output gene regulation across diverse and engineered cell states, we must integrate not only DNA sequence, chromatin state, and RNA levels but also the proteomic network, including weak, transient, or post-translational modification (PTM)-dependent interactions that influence stability, localization, and formation of membraneless organelles containing gene regulators.

Building reliable models will require high-throughput, quantitative measurements on synthetic libraries that vary biophysical parameters one at a time, paired with single-cell readouts of RNA levels, transcription factor binding, and chromatin states. Iterating between such datasets and interpretable machine-learning models will allow us to discover and test gene regulation principles directly and fill in the missing parameter space not captured by natural genomic variation. With these tools, synthetic biology can move from trial and error to true design, unlocking circuits that operate robustly in the full range of cellular states we aim to engineer or treat.

DECLARATION OF INTERESTS

Y.Y.C. has an inventory of patents whose value may be affected by this publication. Y.Y.C. is member of the scientific advisory board of and holds equity in Harton Inc., Lyell Immunopharma, Notch Therapeutics, Pluto Immunotherapeutics, Prime Medicine, Sonoma Biotherapeutics, and Waypoint Bio. J.N.L. is a co-founder and chief scientific officer of Syenex Inc., consults for Arvada Therapeutics, and serves on the advisory board of the Mission Hub for Engineered Genetic Control Systems for Advanced Therapeutics at the University of Edinburgh. L.M. generated IP for the controlling of morphogenetic fields and spatial differentiation through USC. S.R. is a shareholder of Concinnity Genetics Ltd and a member of its scientific advisory board. S.R. is a founder of Zythera Ltd. L.B. is a co-founder of Stylus Medicine and a member of its scientific advisory board.